Quantum information and computation

Quantum circuits

October 11, 2026

Outline

1. Circuits
2. Quantum circuits
3. Inner products
4. Outer products

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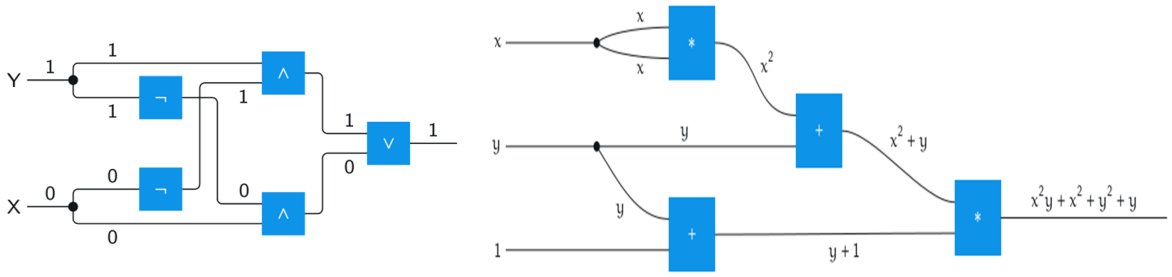
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2 Quantum circuits
3 Inner products
4 Outer products

Circuits

- Circuits provide a convenient model for representing computations.
  - Wires carry information between different parts of the circuit.
  - Gates represent operations that transform the information.
- Circuits are assumed to be *acyclic*, in which information flows in a single direction, typically from left to right, from the input wires to the output wires.
  - Wires store binary values (0 or 1), while gates represent Boolean logic operations that transform these values.
  - Wires carry numerical values, while gates represent arithmetic operations applied to these values.

Circuits

- Circuits are a graphical model used to represent computations. They describe how information flows through a sequence of operations.
- Complex computations can be constructed by combining many gates together, allowing circuits to implement algorithms and computational processes.



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Quantum circuits

- In the quantum circuit model, wires correspond to qubits, while gates represent quantum operations, including both unitary transformations and measurement processes.

$$|0\rangle \qquad H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} \qquad S = \begin{pmatrix}1 & 0\\ 0 & i\end{pmatrix}$$

$$T = \begin{pmatrix}1 & 0\\ 0 & \frac{1+i}{\sqrt{2}}\end{pmatrix} \qquad HSH = \begin{pmatrix}\frac{1+i}{2} & \frac{1-i}{2}\\ \frac{1-i}{2} & \frac{1+i}{2}\end{pmatrix}$$

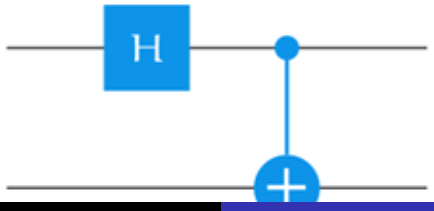


Quantum circuits

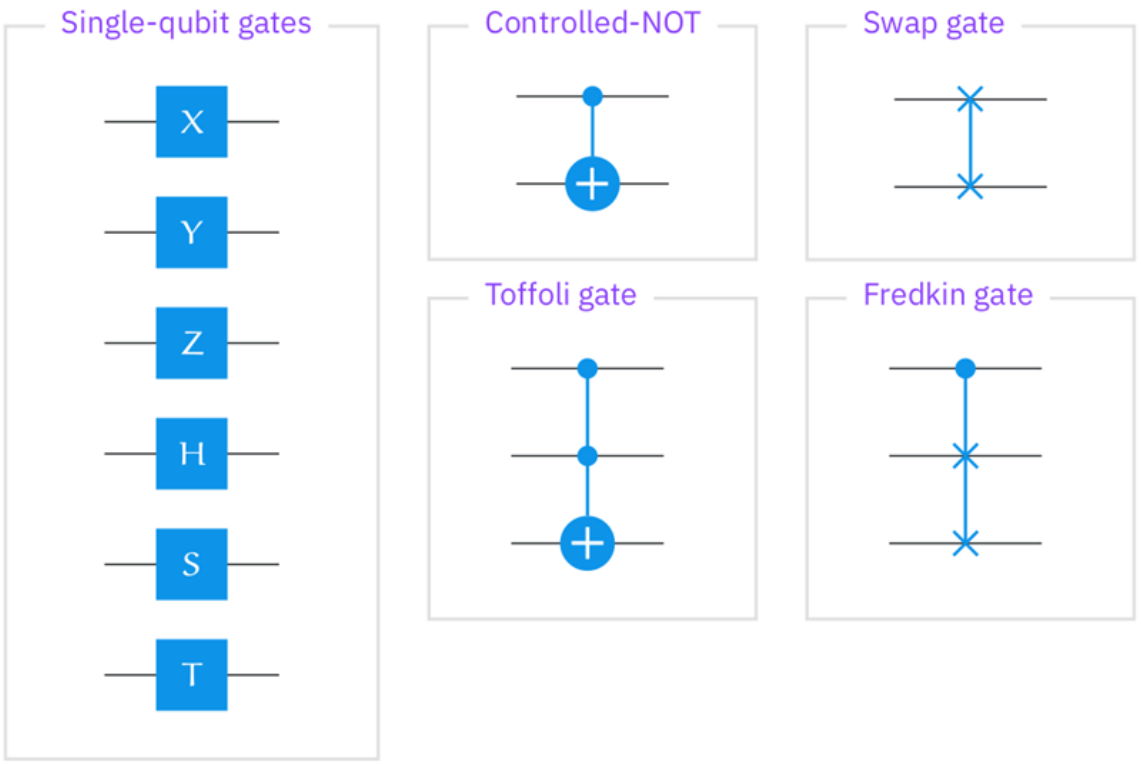
- The CNOT gate with control q_1 and target q_0 is:

$$\text{CNOT} = \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\end{pmatrix}$$

$$U = \text{CNOT} \cdot (H \otimes I) = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0 & 1 & 0\\ 0 & 1 & 0 & 1\\ 0 & 1 & 0 & -1\\ 1 & 0 & -1 & 0\end{pmatrix}$$



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Inner products

- The inner product (also called overlap) between two quantum states, $\langle\psi|$ and $|\psi\rangle$, is a complex number that measures how similar the two states are.
  - Orthogonal states: $\langle\psi|\psi\rangle = 0$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\langle 0|1\rangle = 0$$

  - Identical normalized states: $\langle\psi|\psi\rangle = 1$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$\langle\psi|\psi\rangle = 1$$

Inner products

- A ket is a column vector and its corr. bra is a row vector:

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \quad \langle\psi| = \begin{pmatrix} \alpha_1 & \cdots & \alpha_n \end{pmatrix}$$

- Suppose that we have two kets:

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \qquad |\phi\rangle = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}$$

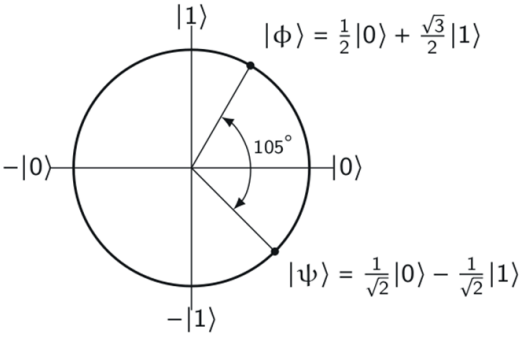
- The inner product of $\langle\psi|$ and $|\phi\rangle$.

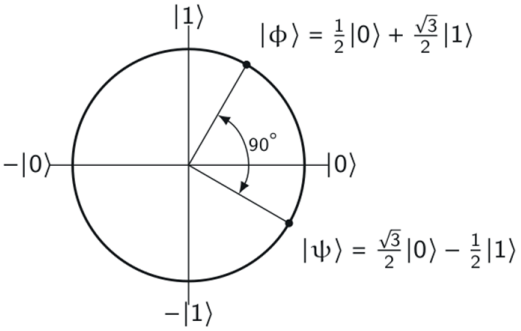
$$\langle\psi|\phi\rangle = \begin{pmatrix} \overline{\alpha_1} & \cdots & \overline{\alpha_n} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \vdots \end{pmatrix} = \overline{\alpha_1}\beta_1 + \cdots + \overline{\alpha_n}\beta_n$$

Inner products

- The I.P of these two vectors is computed as follows, where $\langle 0|0\rangle = 1, \quad \langle 1|1\rangle = 1, \quad \langle 0|1\rangle = \langle 1|0\rangle = 0$:

$$\langle\psi|\phi\rangle = \frac{1-\sqrt{3}}{2\sqrt{2}} = cos(\theta) = cos(105) \approx -0.2588$$





Inner products

- In linear algebra, the **Euclidean norm** (or 2-norm) of a vector

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

is defined as

$$\|\mathbf{v}\| = \sqrt{|v_1|^2 + |v_2|^2 + \cdots + |v_n|^2}.$$

- A quantum state $|\psi\rangle$ corresponds to a column vector:

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}, \quad \langle\psi| = \begin{pmatrix} \overline{\alpha_1} & \overline{\alpha_2} & \cdots & \overline{\alpha_n} \end{pmatrix}.$$

Inner products

- The inner product of the state with itself gives

$$\langle\psi|\psi\rangle = \sum_{i=1}^{n} \overline{\alpha_i}\alpha_i = \sum_{i=1}^{n} |\alpha_i|^2 = \|\psi\|^2.$$

- This shows that the inner product is equivalent to the **squared Euclidean norm**.
  Normalization: A quantum state is normalized if

$$\langle\psi|\psi\rangle = 1 \quad \Longleftrightarrow \quad \|\psi\| = 1,$$

- The sum of measurement probabilities is 1:

$$\sum_i |\alpha_i|^2 = 1.$$

Inner products

- For any two quantum states $|\psi\rangle$ and $|\phi\rangle$

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}, \quad |\phi\rangle = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix},$$

- The inner product satisfies **conjugate symmetry**

$$\langle\psi|\phi\rangle = \sum_{i=1}^{n} \overline{\alpha_i}\beta_i, \quad \langle\phi|\psi\rangle = \sum_{i=1}^{n} \overline{\beta_i}\alpha_i,$$

$$\langle\psi|\phi\rangle = \overline{\langle\phi|\psi\rangle}.$$

- As consequences:
  - If $|\psi\rangle = |\phi\rangle$, then $\langle\psi|\psi\rangle \geq 0$ is real.
  - Swapping the order of the inner product only changes the **complex conjugate**

Inner products

- Two vectors $|\psi\rangle$ and $|\phi\rangle$ are orthogonal if their inner product is zero: $\langle\psi \mid \phi\rangle = 0$.
- An orthogonal set $\{|\psi_1\rangle, \ldots, |\psi_m\rangle\}$ is one where all pairs of vectors are orthogonal:

$$\langle\psi_i \mid \psi_j\rangle = 0 \quad \text{for } i \neq j.$$

- The inner product of the X-basis states $|+\rangle$ and $|-\rangle$:

$$\langle +|-\rangle = \frac{1}{\sqrt{2}}(1 \;\; 1)\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} = \frac{1}{2}(1-1) = 0.$$

- Any two states on opposite sides (e.g., $\langle +|-\rangle$, $\langle i|-i\rangle$) of the Bloch sphere have zero inner product.

Inner products

- The two properties, *normalized* and *orthogonal*, can be combined into a single term: **orthonormal**.
  - $|0\rangle$ and $|1\rangle$, $|+\rangle$ and $|-\rangle$, $|i\rangle$ and $|-i\rangle$ are orthonormal because each state is individually normalized, and they are orthogonal to each other.
  - The Bell basis $\{|\phi^+\rangle, |\phi^-\rangle, |\psi^+\rangle, |\psi^-\rangle\}$ is an orthonormal basis for the two-qubit Hilbert space.
- An orthonormal set $\{|\psi_1\rangle, \ldots, |\psi_m\rangle\}$ is an orthogonal set of unit vectors:

$$\langle\psi_j \mid \psi_k\rangle = \begin{cases}1 & \text{if } j = k,\\ 0 & \text{if } j \neq k,\end{cases} \quad \text{for all } j, k.$$

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Outer products

- Consider two quantum states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\phi\rangle = \gamma|0\rangle + \delta|1\rangle.$$

- The outer product is

$$|\psi\rangle\langle\phi| = (\alpha|0\rangle + \beta|1\rangle)(\gamma^*\langle 0| + \delta^*\langle 1|)$$

$$= \alpha\gamma^*|0\rangle\langle 0| + \alpha\delta^*|0\rangle\langle 1| + \beta\gamma^*|1\rangle\langle 0| + \beta\delta^*|1\rangle\langle 1|$$

- In matrix form:

$$|\psi\rangle\langle\phi| = \begin{pmatrix} \alpha\gamma^* & \alpha\delta^* \\ \beta\gamma^* & \beta\delta^* \end{pmatrix}$$

Outer products

- The outer product of $|\phi\rangle$ and $|\psi\rangle$ is the conjugate transpose of the outer product of $|\psi\rangle$ and $|\phi\rangle$: $|\phi\rangle\langle\psi| = (|\psi\rangle\langle\phi|)^\dagger$.
- Consider the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad |\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

The outer product is

$$|\psi\rangle\langle\phi| = \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right)\left(\frac{1}{\sqrt{2}}(\langle 0| - \langle 1|)\right)$$

$$= \frac{1}{2}(|0\rangle\langle 0| - |0\rangle\langle 1| + |1\rangle\langle 0| - |1\rangle\langle 1|)$$

In matrix form:

$$|\psi\rangle\langle\phi| = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$$

References

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- item Jiri Tomcala, Basic Quantum Computing Algorithms and Their Implementation in Qiskit, Technical University of Ostrava, 2023.